Homework 2

Symbolic Music Generation

b12901045 吳岳霖

Files available at DeepMIR_2026fall/hw2

Outline

- Setup: data, tokenization, models, metrics
- Task A: whole-song generation
  - Results, metrics vs. epoch, song-level spread, structure, longer context
- Task B: continuing an 8-bar prompt
- Bonus: piano accompaniment from a melody
  - Results, chord tokens, why seq2seq fails, sampling sweep
- Future work
- References

Setup: Data and Tokenization

CP packs a bar into about 33 tokens, a third of REMI's 97

Data

- **POP909** training split: 852 songs, 5% held out by song for validation (seed 0)
- Bar lines from **beat_midi.txt**: notes are time-warped so every annotated beat lands on a quarter note
- **Augmentation**, one copy per offset: pitch ±12 semitones, velocity −4 / +5, duration −0.5 / +1 beat; validation never augmented

Tokenizers (miditok, shared settings)

- 88 pitches, 32 velocity bins, 32 tempo bins (40–250 BPM), chord and Program tokens
- 8 positions per beat for notes under 4 beats, 4 per beat up to 12 beats
- **REMI:** one token per event, about **97 tokens per bar**
- **CP (CPWord):** one compound token per note or metrical event, about **33 tokens per bar** (median 2,544 per song)

Setup: Models, Sampling and Metrics

Both models share a GPT-2 backbone; they differ in token representation and output head

	REMI	CP Transformer
Backbone	GPT-2, 12 layers, 8 heads, 512-d	Same
Input	Token embedding	One embedding per token type, concatenated, projected to 512-d
Output	One softmax over the vocabulary	Two stages: Family first, then other types
Context	512 tokens (≈ 5 bars)	512 compound tokens (≈ 16 bars)
Optimiser	AdamW 3e-4, cosine, 1,000 warmup steps	AdamW 3e-4, cosine, 150 warmup steps
Batch size	32	32
Final loss	≈ 0.3	≈ 0.5

Sampling

- Temperature 1.0, top-p 0.95, checkpoints at epochs 1, 3, 5
- **Task A:** 20 songs × 32 bars per checkpoint
- **Task B:** 3 prompts, 8 prompt bars + 24 generated

Metrics

- **H1 / H4:** pitch-class entropy over 1 / 4 bars
- **GS:** groove similarity of bar onset patterns
- **SI short / mid / long:** structureness for 3–8 s, 8–15 s, ≥ 15 s segments
- **Real reference:** 100 random 32-bar POP909 windows

Task A: Results

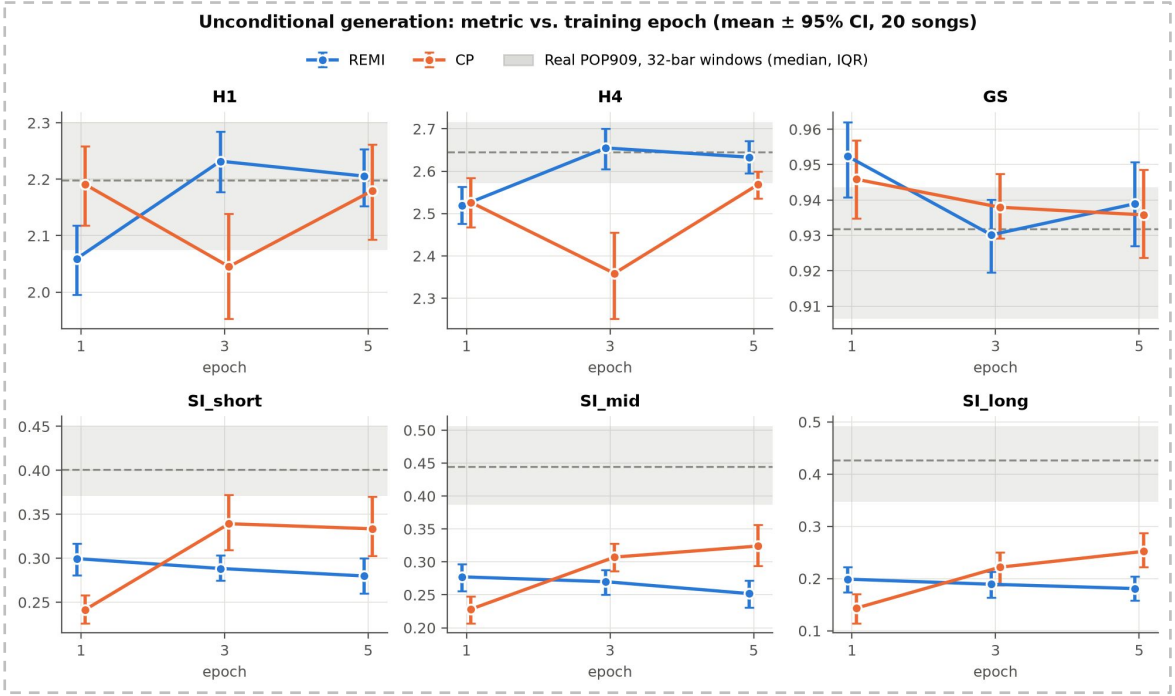
Pitch and rhythm reach real levels by epoch 5; structure stays at 60% of real

Run	loss	H1	H4	GS	SI_short	SI_mid	SI_long	Distance to real
Real (32-bar)		**2.191**	**2.634**	**0.924**	**0.412**	**0.446**	**0.417**	**0**
REMI ep1	1.25	2.059	2.519	0.952	0.300	0.277	0.199	1.24
REMI ep3	0.62	2.232	2.655	0.930	0.288	0.270	0.189	0.96
REMI ep5	0.33	2.206	2.633	0.939	0.280	0.252	0.181	1.01
CP ep1	0.77	2.191	2.527	0.946	0.242	0.228	0.143	1.34
CP ep3	0.62	2.045	2.359	0.938	0.339	0.307	0.222	1.24
CP ep5	0.54	2.180	2.568	0.936	0.334	0.324	0.252	0.75

Means over 20 songs per run. Real = 100 random 32-bar POP909 training windows. Distance to real = mean |z| over the six metrics, z = (run mean − real mean) / real std. Highlighted: closest run to real data.

Task A: Metrics vs. Epoch

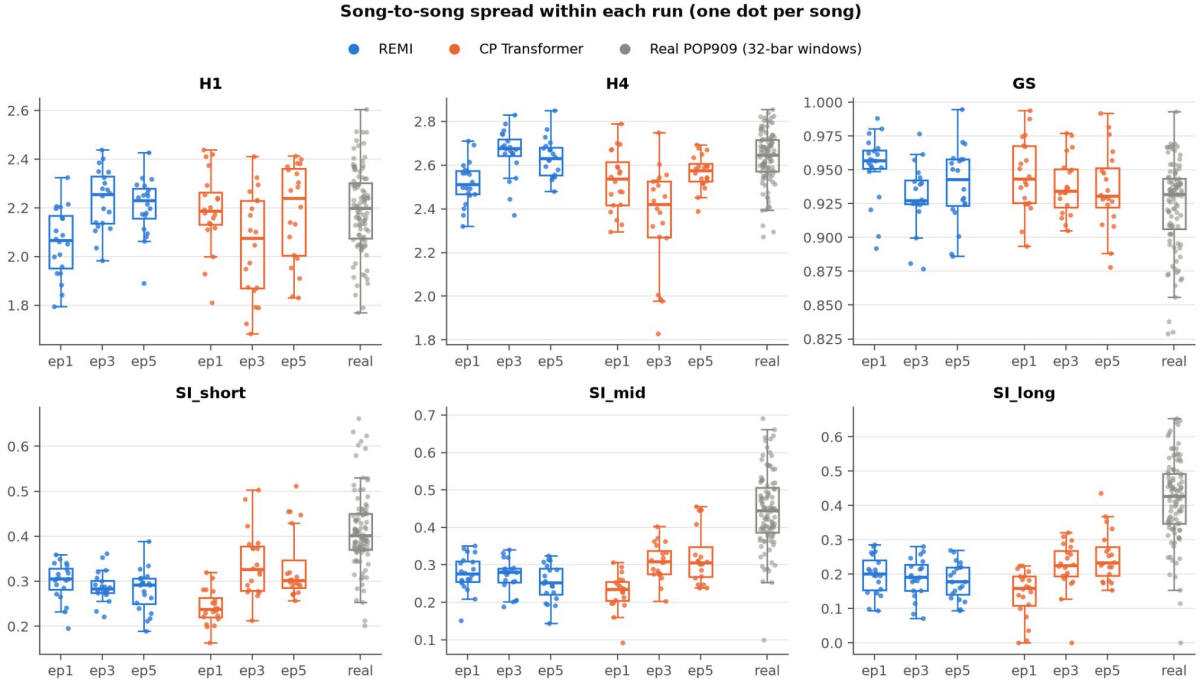
REMI improves pitch statistics with training; CP improves structure



- **REMI:** H1 2.06 → 2.21 and H4 2.52 → 2.63 from epoch 1 to 5 ($p < 0.001$); SI_long flat (0.199 → 0.181, $p = 0.30$)
- **CP:** SI rises at every scale (SI_long 0.143 → 0.252, $p < 0.001$, $d \approx 1.5$); H1, H4, GS flat
- **CP epoch 3 is unstable:** H4 drops to 2.36 as some songs use very few pitch classes
- **Likely reason CP wins on structure:** 512 tokens cover about 16 bars for CP but 5 for REMI

Task A: Song-to-Song Spread

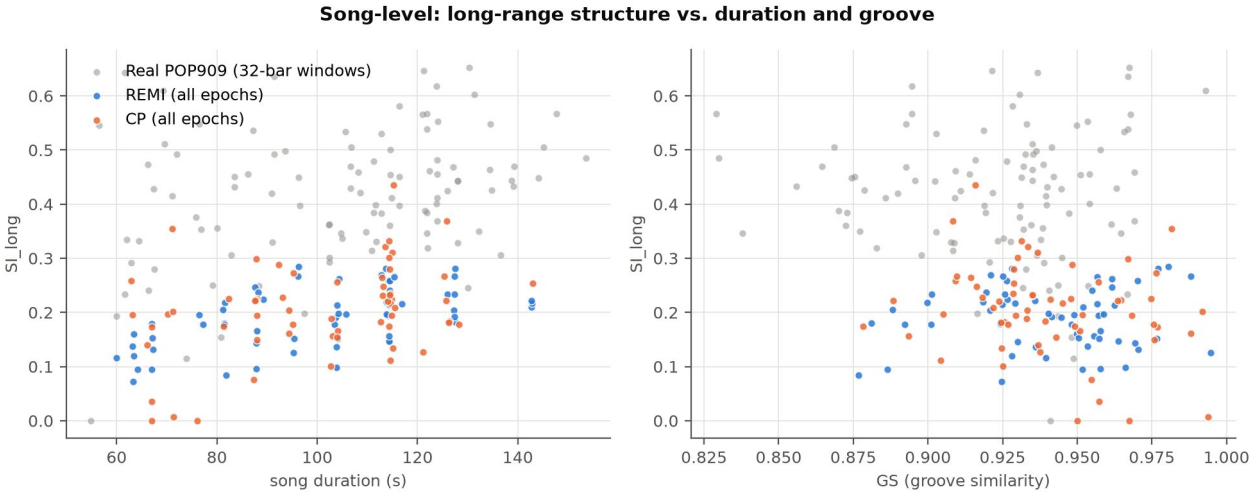
Songs within one run differ about as much as runs do



- REMI ep5 H1 spans **1.89–2.43**; CP ep5 SI_long spans **0.15–0.44**
- With 20 songs, run differences under about 0.5 std are not reliable
- **CP produces the extremes:** songs with no repetition (SI_long = 0) and the most structured song (ep5 #014, 0.435)
- REMI's songs are more uniform

Task A: What Separates Songs

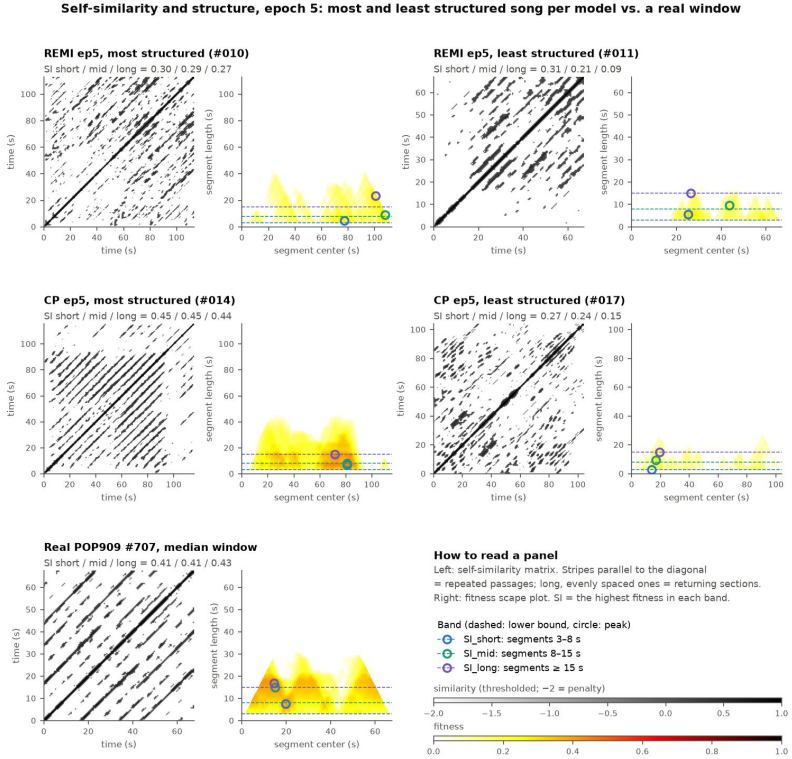
Longer songs score higher SI_long, because SI's bands are measured in seconds



- **Duration:** r = +0.43 with SI_long over 120 songs; at a fixed 32 bars, longer means slower tempo
- Run-level durations are similar (96–108 s vs. 105 s real), so duration does not explain the gap between models
- **Note density:** r = −0.47 with GS; more onsets per bar leave more room for bars to differ

Task A: Structure Up Close

Real songs bring back whole sections; generated songs repeat only short fragments



- **Real music:** long, regularly spaced diagonal stripes in the self-similarity matrix, as whole sections return
- **Generated songs:** short, scattered stripes; fragments of a few bars recur locally, and no section comes back
- Even the most structured generated song (CP ep5 #014, SI_long 0.435) only reaches the real median
- Figure rows: REMI ep5 most / least structured, CP ep5 most / least structured, a median real window

Task B: Results

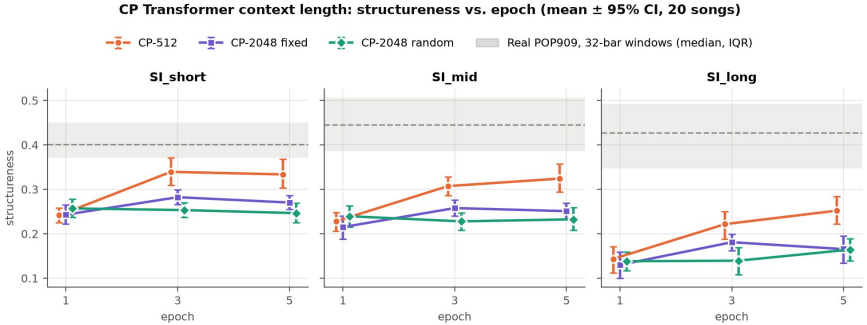
CP keeps the prompt's key better than REMI: 0.87–0.92 vs. 0.75–0.86

Run	H1	H4	GS	SI_short	SI_mid	SI_long	Key consistency
Real (32-bar)	**2.191**	**2.634**	**0.924**	**0.412**	**0.446**	**0.417**	**0.955**
REMI ep1	2.130	2.570	0.949	0.288	0.237	0.169	0.747
REMI ep3	2.290	2.701	0.938	0.332	0.304	0.241	0.858
REMI ep5	2.195	2.620	0.950	0.336	0.291	0.251	0.794
CP ep1	2.208	2.589	0.954	0.328	0.270	0.173	0.870
CP ep3	2.240	2.598	0.956	0.304	0.293	0.253	0.917
CP ep5	2.227	2.617	0.952	0.369	0.341	0.281	0.876
CP-2048 fixed chunk ep5	2.149	2.489	0.954	0.324	0.275	0.198	0.909

Means over the 3 basic-b prompts; metrics cover all 32 bars (8 prompt + 24 generated). Key consistency = cosine similarity of pitch-class histograms, prompt vs. generated bars. Descriptive only.

Task A: Longer Context (CP-2048)

A 4× longer context lowers structureness, with fixed chunks or random crops



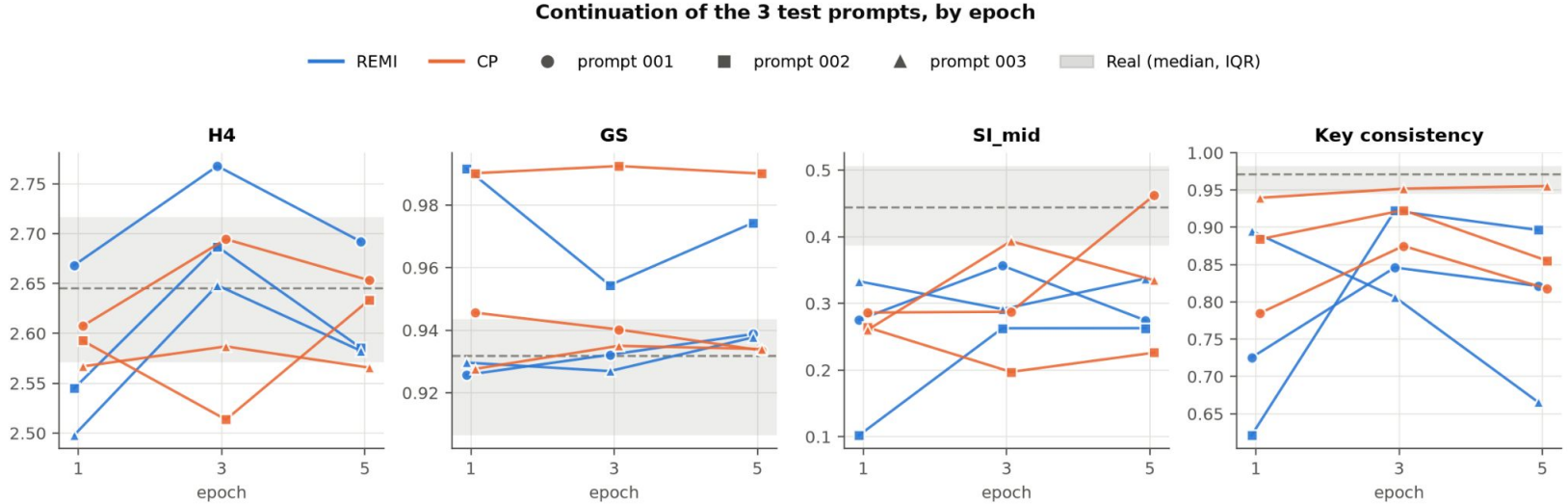
- **Fixed chunk:** same non-overlapping chunks; drops a third of training tokens, including the 139 of 852 songs shorter than 2048 tokens
- **Random:** a new bar-aligned 2048-token window per sample each epoch; short songs padded, same music per epoch as at 512

Run, epoch 5	H1	H4	GS	SI_short	SI_mid	SI_long	Distance to real
CP-512	2.180	2.568	0.936	0.334	0.324	0.252	0.75
CP-2048 fixed chunk	2.202	2.506	0.942	0.270	0.251	0.166	1.24
CP-2048 random	2.088	2.431	0.945	0.246	0.232	0.164	1.53

Why: 512 tokens already cover about 16 bars, and an SI_long segment is about 5 bars. 2048-token batches also get 4× fewer optimizer steps per epoch (batch size 32); epoch-1 H1 of 1.51–1.65 vs. 2.19 shows underfitting.

Task B: Per-Prompt Results

The prompt decides some metrics more than the model does



- **REMI loses the key:** down to 0.62 (ep1, prompt 002) and 0.67 (ep5, prompt 003); with 5 bars of context the prompt drops out of view
- **Prompt 002** gives GS 0.95–0.99 and SI_long 0.10–0.21 in every run: models copy its rhythm without building sections

- **More structured than Task A** (CP ep5 SI_long 0.281 vs. 0.252), partly because the 8 real bars count and can be repeated
- **CP-2048 fixed** keeps the key slightly better (0.909 vs. 0.876) but has lower SI_long and H4, as in Task A

Bonus: Accompaniment Setup

Write a 16-bar piano part for a given melody, scored against the real piano

Task and metrics

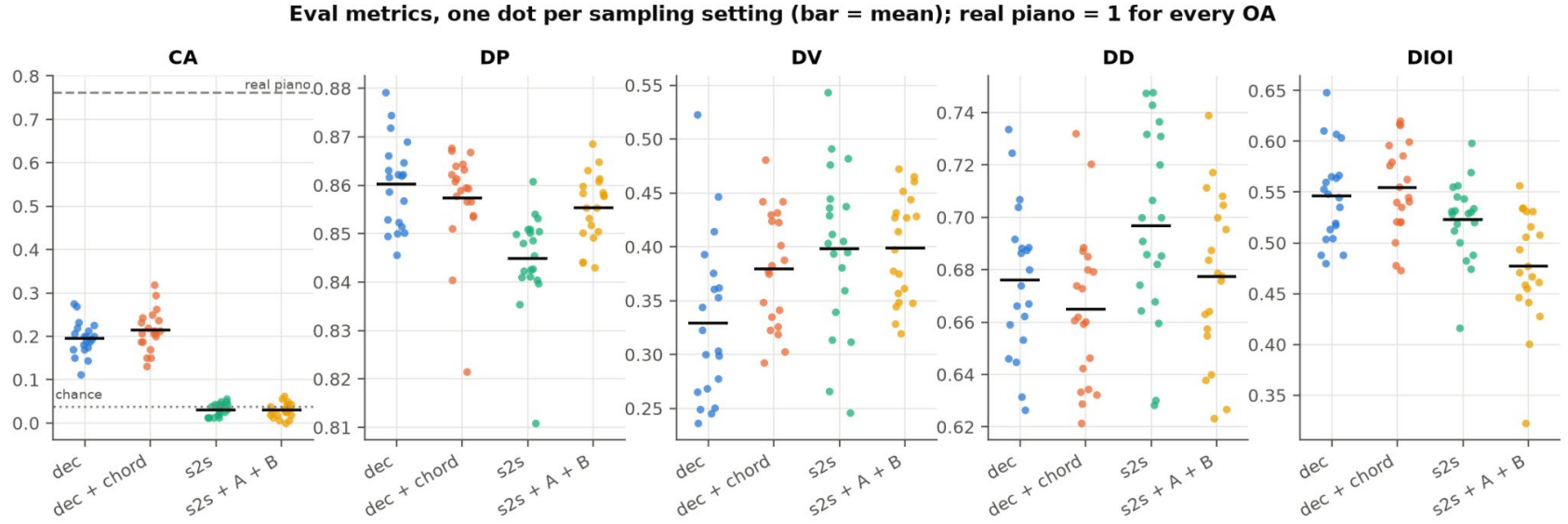
- 10 prompts of 16 melody bars each (160 bars)
- **CA (chord accuracy):** per bar, the piano's best-fit triad vs. the labelled chord; real-piano ceiling **0.762**
- **OA (overlap area, real = 1):** pitch class (DP), velocity (DV), duration (DD), inter-onset interval (DIOI)
- Tokenizing and decoding the real piano gives CA 0.75 and OA 0.96–1.0, so quantization costs almost nothing

Models and sweep

- **dec:** GPT-2, 8 layers, 26.4M params; melody and piano interleaved bar by bar
- **s2s:** BART 4 + 4 layers, 31.7M params; encoder reads all 16 melody bars
- **+ chord:** the model predicts a chord token per bar before its piano
- **+ A + B:** bar-index embedding (A), decoder history dropout at rate *[fill in]* (B)
- AdamW 3e-4, batch 8, 7 epochs *[fill in if changed]*; best checkpoint by validation loss
- **Sweep:** temperature 0.7–1.1 × top_p 0.8–1.0, 20 settings per model

Bonus: Results

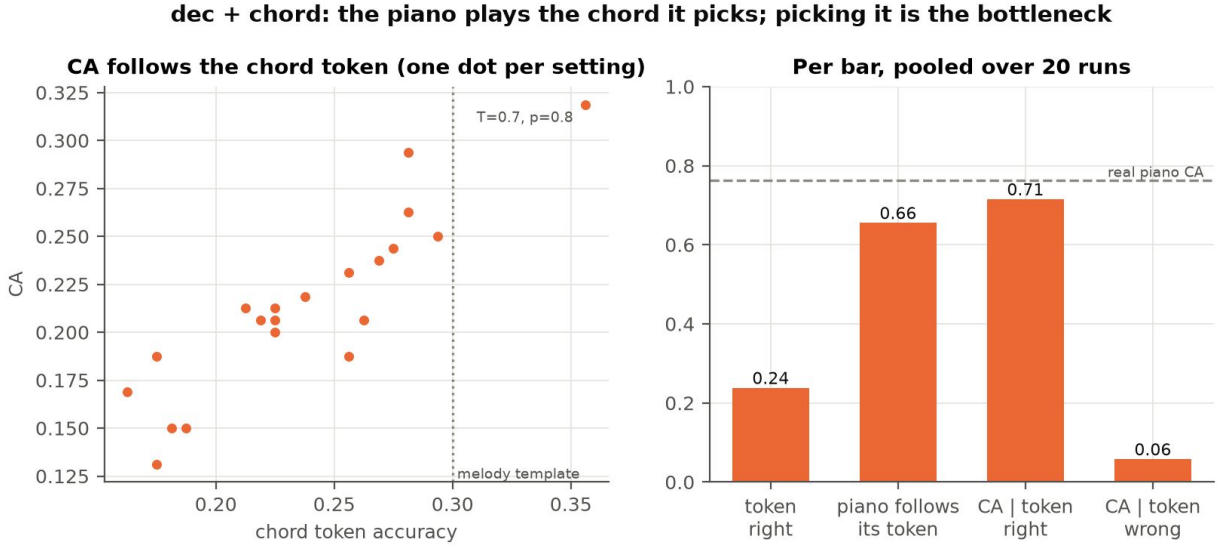
Only the decoder-only model follows the melody; seq2seq stays at chance



- Mean CA over 20 settings: dec **0.195**, dec + chord **0.214** (best 0.319), s2s 0.031, s2s + A + B (explained in next slides) 0.029; chance 0.035–0.041
- OA barely separates the models (DP 0.85–0.86 for all four): matching the real distributions does not require following the melody

Bonus: Chord Tokens

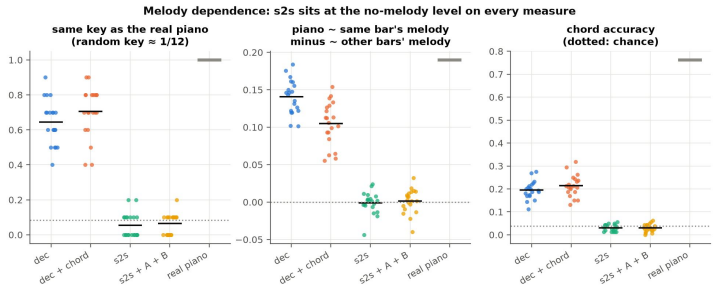
The piano plays the chord token well, but the token is right in only 24% of bars



- CA is **0.715** when the token is right and 0.057 when wrong; across settings CA tracks token accuracy (r = 0.90)
- The token is right in **0.238** of bars, below the melody-only chord template's **0.300**
- Wrong chords are mostly close: root a 4th / 5th away (33%), relative major / minor (19%)
- vs. dec: CA +0.018 (better in 15 of 20 settings, p = 0.10), velocity OA +0.050 (p = 0.004)

Bonus: Why Seq2seq Fails

On all four tests, s2s scores what a model that never reads the melody would



- A+B: indicates adding a bar-embedding and decoder history dropout. In order to make the model more independent of the decoder and the position of the bar more significant.
- **Not a code bug:** validation loss is 1.2779 with the real melody, 1.2827 with another song's, 1.2792 with none
- After a bar or two, the piano history predicts key, rhythm and most chord changes, so the melody adds little to the loss
- In dec, bar i's melody sits right before bar i's piano, so it is as easy to attend to as the history

Model	**Plays in the song's key?** same key as the real piano	**Fits its own song's melody?** own melody minus other songs'	**Fits each bar's melody?** same bar minus other bars	**Plays the labelled chord?** chord accuracy (CA)
No melody used	*≈ 0.08*	*0*	*0*	*≈ 0.04*
Real piano	1.00	0.455	0.190	0.762
dec	0.65	0.452	0.141	0.195
dec + chord	0.71	0.442	0.105	0.214
s2s	0.06	−0.002	−0.001	0.031
s2s + A + B	0.07	−0.032	0.001	0.029

Future Work

Tasks A and B: more structure

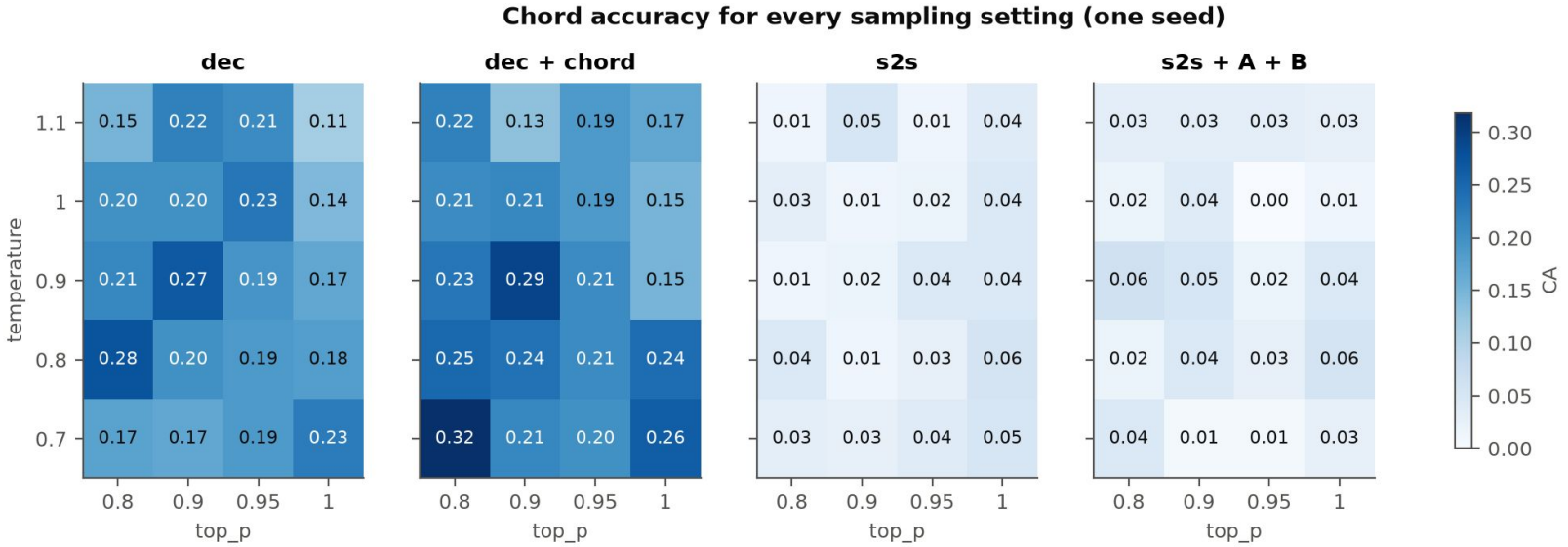
- **Sampling sweep on CP ep5:** lower temperature or top-p, no training needed; temperature 1.0 blurs repeated material
- **Checkpoints by validation loss** instead of a fixed epoch, given how unstable CP ep3 was
- **Drop REMI's Program token:** POP909 uses one program, so it wastes about 1 token in 5
- **Explicit structure tokens** as in Compose & Embellish, which raised SI_long from 17.9 to 25.8 (real 34.8) over a CP baseline

Bonus: better melody conditioning

- **Chord-token sampling (implemented, not yet run):** pick the chord greedily and add a melody-template prior
- **s2s:** history dropout near 0.5, cross-attention limited to the bar's own melody, and real- vs. wrong-melody validation loss logged every epoch

Bonus: Sampling Sweep

The sampling setting explains only 10–14% of CA variance; 73% is sampling noise



- **top_p = 1.0 is worst** for CA (dec 0.166 vs. 0.20–0.21); lower temperature helps (dec + chord 0.248 at 0.7 vs. 0.177 at 1.1)
- **Velocity OA moves the other way** (dec DV 0.28 at top_p 0.8 vs. 0.39 at 1.0): higher CA trades against realistic dynamics

References

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